

UCS638: SECURE CODING

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Course Objectives: This course aims to provide an understanding of the various security attacks and knowledge to recognize and remove common coding errors that lead to vulnerabilities. It gives an outline of the techniques for developing a secure application.

Introduction: Security, CIA Triad, Viruses, Trojans, and Worms, Security Concepts- exploit, threat, vulnerability, risk, attack, Rootkits, Trapdoors, Botnets, Key loggers, Honeypots. Active and Passive Security Attacks.

Need for secure systems: Proactive Security development process, Secure Software Development Cycle (SSDLC), Security issues while writing SRS, Design phase security, Development Phase, Test Phase, Maintenance Phase, Writing Secure Code – Best Practices SD3 (Secure by design, default and deployment), Security principles and Secure Product Development Timeline.

Threat modelling process and its benefits: Identifying the Threats by Using Attack Trees and rating threats using DREAD, Risk Mitigation Techniques and Security Best Practices. Security techniques, authentication, authorization. Defense in Depth and Principle of Least Privilege.

Software & Web Security: Return-to-libc attack, format string vulnerability. Race condition vulnerability, Dirty COW, PE Code injection. Cross site request forgery: CSRF attacks on HTTP GET and POST services & countermeasures. XSS attack: self-propagating XSS worm, preventing XSS attacks, SQL injection attack & countermeasures. Client-side attacks

Laboratory Work:

In this Lab, student shall learn to recognize and remove common coding errors that lead to vulnerabilities. This lab also gives an outline of the techniques for developing a secure application code, implementing different types of attacks and protection schemes for both software and web security. Evaluation will be mainly based on projects and assignments.

Course Learning Outcomes (CLOs) / Course Objectives (COs):

After the completion of the course, the student will be able to:

1. Demonstrate skills needed to deal with common programming errors and develop secure applications.
2. Implement PE Code injection and demonstrate control hijacking via EIP manipulation
3. Demonstrate client-side attacks and identify nature of threats to software and incorporate secure coding practices throughout the planning and development of software product.
4. Demonstrate SQL injection, XSS attack and suggest countermeasures for the same.

Text Books:

1. Howard, M. and LeBlanc, D., Writing Secure Code, Howard, Microsoft Press (2002) 2nd Edition.
2. Deckard, J., Buffer Overflow Attacks: Detect, Exploit, Syngress (2005) 1st Edition.
3. Wenliang Du, Computer Security: A hands-on approach, CreateSpace (2017).

Reference Books:

1. Swiderski, F. and Snyder, W., Threat Modeling, Microsoft Professional, (2004) 1st Edition.
2. Salt, C., J., SQL Injection Attacks and Defence, Elsevier (2012), 2nd Edition.

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST+EST	70
2.	Sessional (May include Assignments//Quizzes)	30